Type MS · Match hormones to glands and effects, and contrast them with nerves

Answer 1 of 9 · Question 1 part whole · 7 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part whole in this mark scheme extract

1				7

Body processes	Substance or structure responsible
regulation of water levels in the blood	ADH
conscious thought	cerebrum
transport of oxygen	haemoglobin
urea formation	liver
control of metabolic rate	thyroxine
detection of body position	semicircular canals
reduction of blood flow to the gut	adrenaline

Total Marks 7

Type MS · Match hormones to glands and effects, and contrast them with nerves

Answer 1 of 9 · Question 1 part whole · 7 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Match hormones to glands and effects, and contrast them with nerves

Answer 2 of 9 · Question 6 part a.ii · 1 mark

2020 NOV · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

6	(a)	(i)	A; (brain)		1
		(ii)	B; (endocrine)		1
	(b)	(i)	• 75 (from weight of man) (1) • 320 × 2 (extra from drinking beer) (1) • 715cm^3 (1)		3
		(ii)	• 715 - 75 (1) • (640 ÷ 75) × 100 (1) • = 853% (1)		3
	(c)		any two from • increases permeability of collecting duct (1) • more water reabsorbed (1) • into blood (1)		2
	(d)		any two from • red blood cells loses water (1) • become distorted (1) • less oxygen transported (1) • volume of plasma decreases (1) • blood becomes thicker/more difficult to circulate(1) • increase in blood pressure (1)	Ignore less water in blood Allow water potential decreases	2
	(e)		any three from • waste products continue to be produced (1) • toxic (1) • need to be removed in urine (1) • less reabsorption of water (1) • less ADH produced (1)	Allow named waste product(s)	3

Total 15 marks

Type MS · Match hormones to glands and effects, and contrast them with nerves

Answer 2 of 9 · Question 6 part a.ii · 1 mark

2020 NOV · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks
		Allow reverse for whole	

Type MS · Match hormones to glands and effects, and contrast them with nerves

Answer 3 of 9 · Question 4 part b.i · 1 mark

2021 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

number				
4 (a)	(i)	• eat too much; • too little exercise; • increased fat storage;		3
	(ii)	• heart disease/attacks/strokes; • causes joints in legs to develop arthritis;	Allow high blood pressure/hypertension	2
(b)	(i)	C (pancreas); A is for excretion B is site of insulin action D produces other hormones		1
	(ii)	B (liver); A is for excretion C is site of production D produces other hormones		1
(c)	(i)	26378 - 22092 = 4286; $\frac{4286 \times 100}{22092}$; = 19.4%;	Allow $\frac{26378 - 22092}{22092}$ for 2 marks Ecf from first calculation max 2 marks	3
	(ii)	0.9 x 3.8 = 3.4(2) million;		1
(d)		any three from • less blood flows; • death of tissues; • increased risk of infection; • less white blood cells to deal with infection;		3

Total 14 marks

Type MS · Match hormones to glands and effects, and contrast them with nerves

Answer 3 of 9 · Question 4 part b.i · 1 mark

2021 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Match hormones to glands and effects, and contrast them with nerves

Answer 4 of 9 · Question 3 part a.iii · 1 mark

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.iii in this mark scheme extract

Number				
3 (a)	(i)	3 320 000;		1
	(ii)	6.00239; x 10^5;	Allow 6.0024	2
	(iii)	pancreas;		1
(b)	(i)	120 ÷ 1.95^2/3.8029; 31.6;	Allow figure not rounded - 31.5581 Allow 31.56/32	2
	(ii)	obese;	ecf from previous question	1
(c)	(i)	• as BMI increases the risk of diabetes increases; • between a BMI of 24-35 the risk of diabetes increases more rapidly/BMI less than 25 the risk increases less rapidly;	Allow any range values between 24-35	2
	(ii)	Any two from the following: • balanced diet; • eat less fat; • eat less sugar	Ignore healthy diet Allow reduce alcohol/carbohydrate consumption for mp3	2

Total marks 11

Type MS · Match hormones to glands and effects, and contrast them with nerves

Answer 4 of 9 · Question 3 part a.iii · 1 mark

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Match hormones to glands and effects, and contrast them with nerves

Answer 5 of 9 · Question 3 part f · 2 marks

2024 JUN · Paper 02 · find part f in this mark scheme extract

Number			
3 (a)	Smoking cigarettes; Causes (lung) cancer/emphysema/bronchitis/heart disease/eq ; OR lack of exercise; obesity/heart disease/high blood pressure/ type ii diabetes/eq;	Allow drugs or named drug and their effect on health for 2 marks.	(2)
(b)	Any two from: • production of bile reduced/affected; • less fat emulsified/fat not emulsified; • lipase less active/lipase unable to digest as much (fat)/slower lipase activity;	Allow bile not produced Allow description of emulsification	(2)
(c)	blurred/misty vision/blindness/less light passing through the lens / less light reaching the retina;		(1)
(d)	cerebrum/cerebellum/ cerebral cortex/cerebral hemispheres/hypothalamus/medulla (oblongata);	Allow any part of the brain that is affected by alcohol	(1)
(e)	Any three from: • reduces release of neurotransmitters; • reduced diffusion across synapse; • less neurotransmitters bind to receptors on post-synaptic membrane; • slows/ fewer transmission of impulses; • slower reaction /reflexes;		(3)
(f)	Any two from • increased heart rate; • increased breathing rate; • pupil dilation; • increase in blood sugar/glucose levels/ increased breakdown of glycogen to glucose; • increase in blood flow to brain/muscles; • increase in (aerobic) respiration (in muscles);		(2)

Total for Question 3 = 11 marks

Type MS · Match hormones to glands and effects, and contrast them with nerves

Answer 6 of 9 · Question 7 part c · 1 mark

2025 JUN · Paper 02 (regional R) · find part c in this mark scheme extract

7	(a)	amylase;		1
	(b)	Any two from • enzymes are immobilised on testing strip; • react with/breaks down glucose (in urine); • (catalyse a chemical reaction that) causes a colour change (in strip);		2
	(c)	pancreas;		1
	(d)	• rejection/immune response/allergies; • (immune system) recognised it as foreign/antigen;		2
	(e)	any two from • use restriction enzymes; • cut out insulin gene from human DNA; • insert human insulin gene into yeast genome/plasmid/genetic material;		2
		Total for Question 7 = 8 marks		

Type MS · Match hormones to glands and effects, and contrast them with nerves

Answer 7 of 9 · Question 2 part b.ii · 2 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

2	(a)	(i)	• A = gall bladder; • B = liver; • C = oesophagus; • D = pancreas;	C- accept pancrease	4
		(ii)	bile;		1
		(iii)	• emulsifies fats/lipids/breaks into small droplets; • increases surface area; • for enzyme/lipase (to act); • neutralises (stomach) acid;		3
	(b)	(i)	any four from • pancreatic juice; • (into th)e small intestine/duodenum; • protease digests/breaks down proteins; • amylase digests/breaks down starch; • lipase digests/breaks down lipids; • (into) maltose/amino acid/glycerol/fatty acid;	If they do not get mp3, mp4, or mp5 They can get a mark for enzyme/named enzyme	4
		(ii)	• insulin; • glucagon;		2
					Total = 14

Type MS · Match hormones to glands and effects, and contrast them with nerves

Answer 7 of 9 · Question 2 part b.ii · 2 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Match hormones to glands and effects, and contrast them with nerves

Answer 8 of 9 · Question 5 part b.i · 2 marks

2020 JAN · Paper 01 · find part b.i in this mark scheme extract

Number				
5 (a)	(i)	12 x 50 = 600 (1) 600 ÷ 1000 = 0.6 (1)	Allow 600 without working out. Allow full marks for correct final answer.	2
	(ii)	provides insulation/prevents nerve impulse dispersing/speeds up nerve impulse (1)		1
(b)	(i)	• hormones are transported in blood (1) • speed of blood flow is determined by heart rate/hormones can only travel at the speed blood is moving/blood flow slower (1)	Allow reverse argument for both marking points	2
	(ii)	• recognition of distance divided by time (1) • $1.0 \div 1.2 \times 10^{2}$ / $1.0 \times 10^{6} \div 1.2 \times 10^{8}$ (1) • 0.0083(3) / 8.3×10^{-3} (s) (1)	Allow full marks for correct final answer	3

Total for Question 5 = 8 marks

Type MS · Match hormones to glands and effects, and contrast them with nerves

Answer 9 of 9 · Question 6 part a · 3 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

Number				
6	(a)	<table><tr><th>Nervous system</th><th>Hormonal system</th></tr><tr><td>nerve fibres/cells;</td><td>blood;</td></tr><tr><td>electrical/electrochemical;</td><td>chemical;</td></tr><tr><td>fast;</td><td>slow;</td></tr></table>	Allow nerves / neurones Reject hormone	3
	(b) (i)	arrow drawn left to right;		1
	(ii)	• neurotransmitter; • released from synapse/synaptic knobs; • diffuses across gap; • bind to receptors on next cell;		4
	(iii)	• neurotransmitter only released at one side of synapse; • can only diffuse in one direction; • from high to low concentration; • receptors only present on post synaptic membrane;		4

Total 12 marks

Type MS · Match hormones to glands and effects, and contrast them with nerves

Answer 9 of 9 · Question 6 part a · 3 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks